Compound Weakly Supervised Clustering

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***Abstract*—Clustering is a fundamental and important step in many image processing tasks, such as face recognition and pedestrian detection. The performance of clustering can be largely enhanced if relevant weak supervision information is appropriately exploited. To achieve this goal, in this paper, we propose the Compound Weakly Supervised Clustering (CSWC) method. Concretely, CSWC incorporates two types of widely available and easily accessed weak supervision information from the label and feature aspects, respectively. To be specific, at the label level, the pairwise constraints are utilized as a kind of typical weak label supervision information. At the feature level, the partial instances collected from multiple perspectives have internal consistency and they are regarded as weak structure supervision information. To achieve a more confident clustering partition, we learn a unified graph with its similarity matrix to incorporate the above two types of weak supervision. On one hand, this similarity matrix is constructed by self-expression across the partial instances collected from multiple perspectives. On the other hand, the pairwise constraints, i.e., must-links and cannot-links, are considered by formulating a regularizer on the similarity matrix. Finally, the clustering results can be directly obtained according to the learned graph, without performing additional clustering techniques. Besides evaluating CSWC on 7 benchmark datasets, we also apply it to the application of face clustering in video data since it has vast application potentiality. Experimental results demonstrate the effectiveness of our algorithm in both incorporating compound weak supervision and identifying faces in real applications.**

***Index Terms*—Compound weak supervision, Constrained clustering, Incomplete view**

I. Introduction

In image processing, clustering is still a fundamental and important research topic [1]–[5] due to the abundance of unlabeled data in many areas, such as face recognition and image segmentation. Conventional clustering methods are designed in a strict unsupervised manner. Practically, there is some relevant prior information available usually. If relevant prior information is appropriately employed, the performance of clustering can be further improved. As a result, utilizing relevant prior information to boost clustering performance, which is named as weakly supervised clustering, has attracted increasing attention.

Commonly, the relevant prior information used in weakly supervised clustering mainly comes from two dimensions: label and feature. In the label level, pairwise constraints, referred to as weak label supervision information in this paper, are frequently used to improve clustering performance. Concretely, pairwise constraints include must-link (ML) and cannot-link (CL) constraints, specifying that the pair of instances must be in the same or different clusters, respectively. In the feature level, for data with multiple consistent and complementary feature representations, the underlying structure of one representation can be seen as prior information for the others [6]–[11]. In such cases, the clustering partition which matches multiple underlying structures most is preferred. When data corruption happens due to equipment failure or human error, the underlying structure may be damaged. As a result, the damaged or incomplete underlying structure is a kind of weakened prior information, and is termed weak structure supervision information in this paper.

Both the weak label supervision information and weak structure supervision information stated above play important roles in increasing clustering confidence, thereby improving clustering performance. Take webpage clustering for illustration. As shown in Fig. 1 (a), there are cannot-links within webpages pairs ①⑥ and ②④, and a must-link within ②③, respectively. The text on webpage ① is missing, and the images on webpage ⑤, ⑥ are absent. On one hand, in view of images, these webpages can be clustered by species (tigers and lions) or image style (natural scene and cartoon). On the other hand, webpages can be grouped by species or gender with only text. As shown in Fig. 1 (b), for images, text, and pairwise constraints, the combination of any two of which helps to increase clustering confidence, but a definite clustering can only be obtained using all of them. In other words, both weak label supervision information and weak structure supervision information can help increase clustering performance, but the combination of them is more effective.

The above example is just one of many real-world scenarios. There are still many scenarios where both weak label supervision information and weak structure supervision information are available in practical applications. In the literature, there are only studies about clustering with a single kind of weak supervision information.

From the label aspect, since the pairwise constraints can be naturally represented by the adjacent matrix, many pairwise constrained graph-based clustering methods were proposed. Lu et al. [12] use the given pairwise constraints to modify the graph, and then getting the clustering results on the similarity graph. However, their method can only effectively deal with the must-link constraints, ignoring the cannot-link constraints. When the number of pairwise constraints is larger than a threshold, Wang et al. [13] can handle the constrained spectral clustering without modifying the similarity matrix. Unlike the two methods mentioned above, in paper [14], they prove that

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This work was partially supported by the NSF of China under Grant No. 61922087, Grant No. 61906201 and Grant No. 62006238, the NSF for Distinguished Young Scholars of Hunan Province under Grant No. 2019JJ20020. the NSF for Young Scholars of Hunan Province under Grant 2020JJ5669.

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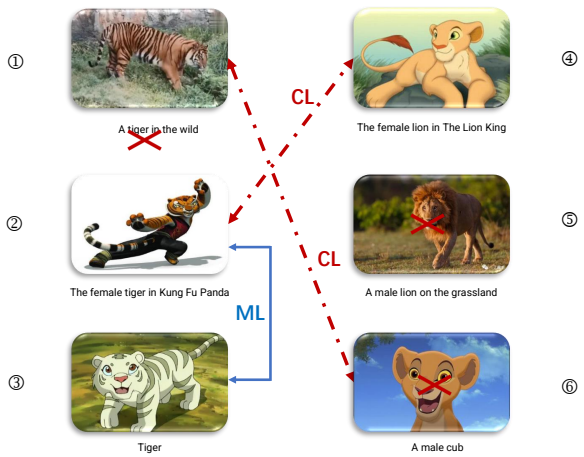


(a)

Criterion / Information	Species (tigers / lions)	Gender (male / female)	Image style (natural scene / cartoon)
Graph	①②③ / ④❺❻	Only use pictures cannot be clustered by gender.	①❺❻ / ②③④
Graph+pairwise constraints	①②③ / ④❺⑥	cannot be clustered by gender.	Since ② and ④ cannot link, clustering by image style is excluded.
Text	❶②③ / ④⑤❻	❶②④ / ❸⑤⑥	Only use text cannot be clustered by gender
Text+pairwise constraints	❶②③ / ④⑤❻	Since ② and ④cannot link, clustering by gender is excluded.	Since ② and ④ cannot link, clustering by image style is excluded.
Text+graph	①②③ / ④⑤❻	❶②④ / ❸⑤⑥	①❺❻ / ②③④
Text+graph+pairwise constraints	①②③ / ④⑤⑥	Since ② and ④ cannot link, clustering by gender is excluded.	Since ② and ④ cannot link, clustering by image style is excluded.

(b)

Fig. 1: There are six webpages containing tigers or lions. Each webpage is represented by images, text, or both. The red cross $\times$ means the corresponding representation is missing. Two must-link samples are linked with a blue solid line, and the cannot-link samples are indicated with red dotted lines. In the table on the right side, a red straight line is used to divide the six webpages into two categories. The numbers with white or black background indicate that the corresponding webpages can be or cannot be confidently grouped, respectively.

the cannot-link constraints can be transformed into a graph regularization term, and utilize the label propagation to solve the constraint.

From the feature aspect, research results on clustering with weak structure supervision information, mainly come from Incomplete Multi-view Clustering (IMC), where data with multiple feature representations is called multi-view data. Existing IMC methods can be roughly divided into two categories: matrix factorization based [15]–[20] and graph-based [21]–[23] methods. Matrix factorization based methods have the ability to reconstruct missing features, but are not optimal for data which is not linearly separable. Graph-based IMC approaches are more applicable for nonlinear cases, but lacking feature reconstruction capability is the fly in their ointment. It can be found that IMC develops along the above two lines but ignores the deep combination between them, except for adding graph regularization to matrix factorization based methods [17].

These methods have achieved prominent performances, but they can not be utilized to solve our problem directly since they are mainly focused on one kind of weak supervision. Though they become applicable after simple modifications, they are still not able to make full use of two kinds of weak supervision information simultaneously, as our experimental results show.

In this paper, to effectively clustering data with multiple incomplete representations and pairwise constraints, we propose the Compound Weakly Supervised Clustering (CWSC) method. On the whole, we dexterously design a graph-based model, which unifies the weak label and weak structure information in a compact form. Specifically, by transforming the pairwise constraints into a regularizer on the graph Laplacian, the validation of pairwise constraints is provable theoretically. Then, by representing the reconstructed data matrices of each view simultaneously with self-expression and matrix factorization techniques, the weak structure supervision information and weak label supervision information are incorporated naturally. Different from the previous methods that directly add a graph regularizer in the matrix factorization model, this trick overcomes the shortcomings of both types of IMC methods. Finally, by adding rank constraints on the graph which contains two kinds of weak supervision information, the clustering results can be obtained directly according to the disconnected components.

In summary, the contribution of this paper can be summarized as follows.

- As far as we know, this study presents the first attempt to solve the coexistence problem of weak structure supervision information and weak label supervision information.
- A unified model is dexterously designed to learn a graph that distills the weak supervision information from both the label and feature aspects. Furthermore, the clustering results can be obtained directly according to the disconnected components of the learned graph.
- With the help of self-expression technique, a byproduct of our algorithm is the combination of matrix factorization based method and graph-based method. It joins the advantages of two types of weakly structure-supervised clustering methods, which are investigated separately in the previous literature.
- An algorithm with proved equivalence is proposed to address the solution difficulty brought by the merge of the self-expression and matrix factorization representations. Systematically experimental results on 7 real-world datasets and face clustering tasks verify the effectiveness of the proposed CWSC method.

II. Notations and Related Works

In this section, we first introduce some basic definitions of symbols. Then we will introduce some related works.

A. Notations

Matrices and vectors will be represented by boldface uppercase and boldface lowercase letters respectively. A n-dimensional vector of all ones is denoted as $\mathbf{1} \in \mathbb{R}^n$. For matrix $\mathbf{A}$, $\mathbf{a}_j$ represents the j-th column of $\mathbf{A}$ and $a_{i,j}$ is the value of the i-th row and the j-th column of $\mathbf{A}$. $\mathbf{A}^T$ and $\|\mathbf{A}\|_F$ represent the Frobenius norm and transpose of $\mathbf{A}$ respectively. The L_1 norm is defined as $\|\mathbf{A}\|_1 = \sum_{i=1}^{m}\sum_{i=1}^{n}|a_{i,j}|$. Suppose the dataset has V views, let $\mathbf{X}^{(v)} = [\mathbf{x}_1^{(v)}, \mathbf{x}_2^{(v)}, \ldots, \mathbf{x}_n^{(v)}] \in \mathbb{R}^{d^{(v)}\times n}$ be the incomplete data matrix, where n is the number of samples, and $d^{(v)}$ is the dimensionality of the v-th view ($v = 1, 2, \cdots, V$). $\{\mathbf{Z}^{(v)}\}_{v=1}^{V} \in \mathbb{R}^{d^{(v)}\times n}$ denote the reconstructed data matrices. An indicator matrix $\mathbf{\Omega}^{(v)} \in \{0,1\}^{d^{(v)}\times n}$ is introduced to record the absence state of elements in $\mathbf{X}^{(v)}$. Specifically, $\mathbf{\Omega}_{i,j}^{(v)} = 0$ indicates that the (i,j)-th element of $\mathbf{X}^{(v)}$ is missing, and 1 otherwise, $\odot$ represents the multiplication of the corresponding elements of the matrices. diag$(\cdot)$ is an operator to change a vector into a diagonal matrix by placing the vector elements on the main diagonal. For a dataset with n samples, $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, we use $\mathcal{M}$ and $\mathcal{C}$ to collect the must-link and cannot-link constraints, respectively. That is

$$\begin{aligned} \mathcal{M} &= \{(\mathbf{x}_i, \mathbf{x}_j) | \forall i, \forall j \geq i, \quad \mathbf{x}_i \quad \text{must} \quad \text{link} \quad \mathbf{x}_j\} \\ \mathcal{C} &= \{(\mathbf{x}_i, \mathbf{x}_j) | \forall i, \forall j \geq i, \quad \mathbf{x}_i \quad \text{cannot} \quad \text{link} \quad \mathbf{x}_j\}. \end{aligned} \tag{1}$$

The notations used in this paper are summarized in Table I.

TABLE I: Notations

Notations	Descriptions
d	The total dimensionality of data
$d^{(v)}$	The dimensionality of the v-th view
n	The data size
r	The dimensionality of subspace
p	The percentage of missing data
c	The number of classes
$\mathbf{x}_i$	The i-th sample
$\mathbf{y} \in \mathbb{R}^n$	There exists a cannot-link indicator constraint between $\mathbf{x}_i$ and $\mathbf{x}_j$, $y_i = 1$, $y_j = -1$
V	The number of views
$\mathbf{X}^{(v)} \in \mathbb{R}^{d^{(v)}\times n}$	The data matrix of the v-th view
$\mathbf{\Omega}^{(v)} \in (0,1)^{d^{(v)}\times n}$	The indicator matrix of $\mathbf{X}^{(v)}$
$\mathbf{Z}^{(v)} \in \mathbb{R}^{d^{(v)}\times n}$	The constructed data matrix of the v-th view
$\mathbf{S} \in \mathbb{R}^{n\times n}$	The similarity matrix, where $s_{i,j}>0$
$\mathbf{U}^{(v)} \in \mathbb{R}^{d^{(v)}\times r}$	The projection matrix in the v-th view
$\mathbf{W} \in \mathbb{R}^{r\times n}$	The latent representation
$\mathcal{M}$	The set of must-link constraints
$\mathcal{C}$	The set of cannnot-link constraints

B. Related Works

In this subsection, we review some related works on clustering with pairwise constraints and Incomplete Multi-view Clustering (IMC), respectively.

For clustering with pairwise constraints, Nie et al. [14] put forward Semi-supervised Clustering via Pairwise Constrained Optimal Graph (PCOG), which is a breakthrough in dealing with the non-transitivity of cannot-link constraints. Specifically, they transformed the cannot-link constraints into a graph regularization term, which theoretically ensures the instances constrained by the cannot-link to be grouped into different clusters by PCOG. Moreover, they design an inquiry strategy for actively selecting pairwise constraints, which effectively improves the clustering performance.

As for IMC, matrix factorization based methods can be summarized into a typical framework [20], [24]

$$\begin{aligned} \min_{\mathbf{U}^{(v)},\mathbf{Z}^{(v)},\mathbf{W}} & \sum_{v=1}^{V} \left\|\mathbf{Z}^{(v)} - \mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 + \gamma\mathcal{R}_1(\mathbf{U}^{(v)}) + \alpha\mathcal{R}_2(\mathbf{W}) \\ \text{s.t.} \quad & \mathbf{\Omega}^{(v)} \odot (\mathbf{Z}^{(v)} - \mathbf{X}^{(v)}) = \mathbf{0}, v = 1, \ldots, V, \end{aligned} \tag{2}$$

where $\mathbf{U}^{(v)} \in \mathbb{R}^{d^{(v)}\times r}$ is the basis matrix, $\mathbf{W} \in \mathbb{R}^{r\times n}$ is the common latent subspace, $\mathcal{R}_1(\mathbf{U}^{(v)})$ and $\mathcal{R}_2(\mathbf{W})$ are regularization terms imposed on $\mathbf{U}^{(v)}$ and $\mathbf{W}$, γ and α are positive trade-off parameters. We can change the regularization terms $\mathcal{R}_1(\mathbf{U}^{(v)})$ and $\mathcal{R}_2(\mathbf{W})$ to design different algorithms in different circumstances [20]. When $\mathcal{R}_2(\mathbf{W}) = \|\mathbf{W}\|_1$, Eq. (2) is the formulation of Partial multi-View Clustering (PVC) [15], while $\mathcal{R}_1(\mathbf{U}^{(v)}) = \left\|\mathbf{U}^{(v)}\right\|_F^2$, and $\mathcal{R}_2$ is a Laplacian graph regularization, Eq. (2) corresponds to the formulation of Incomplete Multi-modality Grouping (IMG) [17].

For graph-based methods, it is tough to represent them with an exact framework. However, most graph-based methods first input the missing multiple-view data, and then use some strategies to improve initialization, or add some constraints to improve the quality of the graph. An representative work is Simultaneous Representation Learning and Clustering (SRLC) [21]. Concretely, SRLC uses the similarity matrix to measure the similarity between two samples. Moreover, SRLC learns the low dimensional representation and common probability label matrix from each view, so it can use the learned common representation and get the clustering results from the probability label matrix to reflect the nonlinear relationship.

III. Formulation

In this section, we provide a solution to the problem of clustering with both weak label supervision information and weak structure supervision information, which is never or seldom considered in the literature. To solve this complicated problem, we first pay attention to its two sub-problems: weakly label-supervised clustering and weakly structure-supervised clustering.

For existing weakly label-supervised clustering methods, the weak structure supervision information is not considered. These methods are probably not sufficient in dealing with incomplete data from multiple views. For weakly structure supervised clustering, we find that the two main categories of methods have some shortcomings. Although the method based on matrix factorization can complete missing data, it cannot handle nonlinear data well. The graph-based methods can make better use of the spatial structure information of the samples, but lack in completing the missing data. All the weakly structure supervised clustering methods ignore the weak label supervision information. Therefore, they are not suitable for processing composite scenes containing two kinds of weak supervision information.

In the process of solving the problem, we encounter the following two main challenges.

Challenge 1 Reasonably describe the weak label supervision information in the model, so that the pairwise constraints can effectively improve the clustering results.
Challenge 2 Design a unified model, which can not only integrate weak label supervision information and weak structure supervision information elegantly, but also solve the problem of weak structure supervised clustering analyzed above.

Since compound weakly supervised clustering is complicated, we solve the above challenges one by one.

In the actual data collection process, the pair-constrained weak label supervision information is easy to obtain. But it is not easy to portray this kind of information in the clustering model. Compared with other methods, it is more intuitive to describe the relationship of the pairwise constraints in the similarity matrix. For two samples $\mathbf{x}_i$ and $\mathbf{x}_j$, if $(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{M}$, we set $s_{i,j}$ and $s_{j,i}$ equal to ι (ι>0), they must belong to the same connected component in the similarity matrix $\mathbf{S}$, but when $(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{C}$, it is almost useless to simply cut off the connection between them, since they are likely to be indirectly connected through other samples. Therefore, importing the cannot-link constraints into the model is a hard problem. Inspired by [14], the cannot-link situation can be expressed as follows.

Theorem 1 ([14]). *For a nonnegative similarity matrix* $\mathbf{S}$*, denote* $\mathbf{y} \in \mathbb{R}^n$ *as the indicator vector of cannot-link pairs. For two samples* $(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{C}$*,* $y_i = 1, y_j = -1$*, and the other elements in* $\mathbf{y}$ *are randomly selected in* $(-1, 1)$ *. When*

$$\mathbf{y}^T \mathbf{L}_{\mathbf{S}} \mathbf{y} = 0, \tag{3}$$

$\mathbf{x}_i$ *and* $\mathbf{x}_j$ *belong to different connected components of* $\mathbf{S}$*, where* $\mathbf{L}_{\mathbf{S}} = \mathbf{D} - \mathbf{S}$ *is the Laplacian matrix of* $\mathbf{S}$*,* $\mathbf{D} = \mathrm{diag}(\mathbf{d})$*,* $\mathbf{d} \in \mathbb{R}^n$*, and* $d_i = \sum_j s_{i,j}$*.*

In this way, the CL constraints are transformed into a graph regularizer. The first challenge is overcome.

From the above, we have established a connection between the weak label supervision information and the similarity matrix. To address the second challenge, a natural idea is to combine the graph-based IMC method with Eq (3). However, the solution can be further improved. In real life, many scenarios have demand of completing the missing data. In order to further expand the application scenarios, we hope that the model has the function of completing the absent data.

As we analyzed earlier, many studies have shown that the matrix factorization based IMC methods can effectively complete the missing data. Therefore, if CWSC can combine the advantages of the two mainstream IMC methods, the model could be better. However, the matrix factorization based IMC method aims to learn a common representation and the graph-based IMC method is dedicated to learning a similarity matrix. There are few similarities between the two types of IMC methods, thus, combing them is not easy.

To find the connection of matrix factorization based method and graph-based method. We believe that the most critical point is to find the direct coupling between original feature matrices and graphs. Fortunately, self-expression technology fits this perfectly.

The main idea of self-expression theory is that each sample can be effectively represented by the combination of other samples [25]. Specifically, for each instance $\mathbf{z}_i \in \mathbb{R}^{d^{(v)}}$ can be written as

$$\mathbf{z}_i^{(v)} = \mathbf{Z}^{(v)} \mathbf{s}_i, \quad \text{s.t.} \quad \mathbf{S1} = \mathbf{1}, \quad \mathbf{S} \geq \mathbf{0}, \tag{4}$$

where $\mathbf{s}_i = [s_{ii}, s_{i2}, \ldots, s_{in}]^T$, s_{ii} is set to zero to avoid the trivial solution of a linear combination of itself. In other words, data matrix $\mathbf{Z}^{(v)}$ is equivalent to a self-expression dictionary, in which each data point can be represented by a linear combination of some other data points. Usually, such a linear representation is usually not unique. Notably, the non-zero elements in $\mathbf{s}_i$ correspond to the sample points in the linear combination of $\mathbf{z}_i^{(v)}$. Inspired by the mentioned framework (2), $\left\|\mathbf{Z}^{(v)} - \mathbf{U}^{(v)}\mathbf{W}\right\|_F^2$ is adopted as the loss function. By introducing (4) into the loss function, we have

$$\begin{aligned} \min_{\mathbf{U}^{(v)}, \mathbf{Z}^{(v)}, \mathbf{W}, \mathbf{S}} \sum_{v=1}^{V} \left\{ \left\|\mathbf{Z}^{(v)} - \mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 + \xi \left\|\mathbf{Z}^{(v)} - \mathbf{Z}^{(v)}\mathbf{S}\right\|_F^2 \right\} \\ \text{s.t.} \quad \mathbf{\Omega}^{(v)} \odot (\mathbf{Z}^{(v)} - \mathbf{X}^{(v)}) = \mathbf{0}, v = 1, \ldots, V, \\ \mathbf{S1} = \mathbf{1}, \quad \mathbf{S} \geq \mathbf{0}. \end{aligned} \tag{5}$$

In this case, we naturally introduce the graph $\mathbf{S}$ in the matrix factorization based model. However, the above loss function is a little complicated. In order to simplify the loss function and reduce the number of parameters, we merge the first two items of the above formula. Then we have

$$\begin{aligned} \min_{\mathbf{U}^{(v)}, \mathbf{Z}^{(v)}, \mathbf{W}, \mathbf{S}} \sum_{v=1}^{V} \left\|\mathbf{Z}^{(v)}\mathbf{S} - \mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 \\ \text{s.t.} \quad \mathbf{\Omega}^{(v)} \odot (\mathbf{Z}^{(v)} - \mathbf{X}^{(v)}) = \mathbf{0}, v = 1, \ldots, V, \\ \mathbf{S1} = \mathbf{1}, \quad \mathbf{S} \geq \mathbf{0}. \end{aligned} \tag{6}$$

To capture the global structure, a graph regularizer $\mathcal{R}(\mathbf{W}) = tr(\mathbf{W}\mathbf{L}_{\mathbf{S}}\mathbf{W}^T)$ is imported. To limit the freedom degree, the orthogonal constraint on $\mathbf{W}$ is imported. Thus, we have

$$\begin{aligned} \min_{\mathbf{U}^{(v)}, \mathbf{Z}^{(v)}, \mathbf{W}, \mathbf{S}} \sum_{v=1}^{V} \left\|\mathbf{Z}^{(v)}\mathbf{S} - \mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 + \alpha tr(\mathbf{W}\mathbf{L}_{\mathbf{S}}\mathbf{W}^T) \\ \text{s.t.} \quad \mathbf{\Omega}^{(v)} \odot (\mathbf{Z}^{(v)} - \mathbf{X}^{(v)}) = \mathbf{0}, v = 1, \ldots, V, \\ \mathbf{W}\mathbf{W}^T = \mathbf{I}. \quad \mathbf{S1} = \mathbf{1}, \quad \mathbf{S} \geq \mathbf{0}. \end{aligned} \tag{7}$$

So far, we successfully combine the graph-based method with the matrix factorization based method, and the missing data matrix $\mathbf{Z}^{(v)}$ and the graph $\mathbf{S}$ are coupled through self-expression. While using the learned common representation to complete the missing data matrix $\mathbf{Z}^{(v)}$, the similarity matrix $\mathbf{S}$ extracts the spatial structure information of the samples.

By solving Eq. (7), we can learn the common representation $\mathbf{W}$ and the similarity matrix $\mathbf{S}$ at the same time, and we can perform clustering on $\mathbf{W}$ or $\mathbf{S}$. Since utilizing $\mathbf{S}$ is easier to combine with the weak label supervision information, in this paper, we obtain clustering results from $\mathbf{S}$.

Suppose there are g_n pairs of CL constraints, according to **Theorem 1**, by combining Eq. (7) and Eq. (3), we have

$$\begin{aligned}\min_{\mathbf{U}^{(v)},\mathbf{Z}^{(v)},\mathbf{W},\mathbf{Y}\in\Psi,\mathbf{S}\in\Phi} &\sum_{v=1}^{V}\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 \\ &+\alpha tr(\mathbf{W}\mathbf{L}_\mathbf{S}\mathbf{W}^T)+\beta\sum_{g=1}^{g_n}\mathbf{y}_g^T\mathbf{L}_\mathbf{S}\mathbf{y}_g \\ \text{s.t.}\quad &\mathbf{\Omega}^{(v)}\odot(\mathbf{Z}^{(v)}-\mathbf{X}^{(v)})=\mathbf{0}, v=1,\dots,V, \\ &\mathbf{W}\mathbf{W}^T=\mathbf{I},\quad \mathbf{S}\mathbf{1}=\mathbf{1},\quad \mathbf{S}\geq\mathbf{0},\end{aligned} \tag{8}$$

where Ψ is the feasible zone for $\mathbf{Y}=[\mathbf{y}_1,\cdots,\mathbf{y}_{g_n}]\in\mathbb{R}^{n\times g_n}$. For the g-th CL pair $(\mathbf{x}_{i_g},\mathbf{x}_{j_g})\in\mathcal{C}$, $\mathbf{y}_g(i_g)=1,\mathbf{y}_g(j_g)=-1$, and the other elements of $\mathbf{y}_g$ are randomly given within $(-1,1)$. Φ is the feasible region of $\mathbf{S}$, i.e., $\Phi=\{\mathbf{S}|\forall(\mathbf{x}_i,\mathbf{x}_j)\in\mathcal{M}:s_{i,j}=s_{j,i}=\iota\}$.

After solving Eq. (8), the learned similarity matrix $\mathbf{S}$ contains two kinds of weak supervision information, in the meanwhile, the absent data matrix can be completed by the learned $\mathbf{W}$.

However, similar as most of previous methods. CWSC still needs to learn a subspace first, and then utilizes k-means or other methods to perform clustering. This two-step strategy separates the clustering from the representation learning, so that finding the global optimal solution becomes harder. Considering that our model introduces the similarity matrix $\mathbf{S}$, obtaining clustering results in one step becomes possible by combining graph-related techniques. Especially, there is an important property of Laplacian matrix [26]. The number of zeros in the eigenvalues of the Laplacian matrix $\mathbf{L}_\mathbf{S}$ is equal to the number of connected components of the similar matrix $\mathbf{S}$ [27]. Inspired by the above theory, we add a rank constraint to the Laplacian matrix $\mathbf{L}_\mathbf{S}$ and arrive at

$$\begin{aligned}\min_{\mathbf{U}^{(v)},\mathbf{Z}^{(v)},\mathbf{W},\mathbf{Y}\in\Psi,\mathbf{S}\in\Phi} &\sum_{v=1}^{V}\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 \\ &+\alpha tr(\mathbf{W}\mathbf{L}_\mathbf{S}\mathbf{W}^T)+\beta\sum_{g=1}^{g_n}\mathbf{y}_g^T\mathbf{L}_\mathbf{S}\mathbf{y}_g \\ \text{s.t.}\quad &\mathbf{\Omega}^{(v)}\odot(\mathbf{Z}^{(v)}-\mathbf{X}^{(v)})=\mathbf{0}, v=1,\dots,V, \mathbf{W}\mathbf{W}^T=\mathbf{I}, \\ &\text{rank}(\mathbf{L}_\mathbf{S})=n-c,\quad \mathbf{S}\mathbf{1}=\mathbf{1},\quad \mathbf{S}\geq\mathbf{0}.\end{aligned} \tag{9}$$

Different from the traditional spectral clustering, after adding the rank constraint, CWSC not only learns an indicator matrix, but also learns a new affinity matrix $\mathbf{S}$. The clustering results can be obtained from $\mathbf{S}$ according to the disconnected components directly, without performing extra k-mean clustering.

IV. Solution and Analysis

In this section, we first introduce the solving and updating rules of several main variables, then we give some technical details analysis and time complexity analysis.

A. The Optimization Procedures

Since we have to optimize six groups of variables and it is very difficult to solve them at the same time, we adopt the alternating iterative strategy.

Denote $\sigma_j(\mathbf{L}_\mathbf{S})$ as the j-th smallest eigenvalue of Laplacian $\mathbf{L}_\mathbf{S}$, and $\mathbf{L}_\mathbf{S}$ is positive semi-definite, the constraint $\text{rank}(\mathbf{L}_\mathbf{S})=n-c$ ensures $\sum_{j=1}^{c}\sigma_j(\mathbf{L}_\mathbf{S})=0$, according to Ky Fan theory [28], we have

$$\sum_{j=1}^{c}\sigma_j(\mathbf{L}_\mathbf{S})=\min_{\mathbf{F}\in\mathbb{R}^{c\times n},\mathbf{F}\mathbf{F}^T=\mathbf{I}}tr(\mathbf{F}\mathbf{L}_\mathbf{S}\mathbf{F}^T). \tag{10}$$

When λ is large enough, the problem in (9) is equivalent to

$$\begin{aligned}\min_{\substack{\mathbf{W},\mathbf{Z}^{(v)},\mathbf{U}^{(v)}\\ \mathbf{S},\mathbf{Y}\in\Psi}} &\sum_{v=1}^{V}\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2+\alpha tr(\mathbf{W}\mathbf{L}_\mathbf{S}\mathbf{W}^T) \\ &+\beta tr(\mathbf{Y}^T\mathbf{L}_\mathbf{S}\mathbf{Y})+\lambda tr(\mathbf{F}\mathbf{L}_\mathbf{S}\mathbf{F}^T) \\ \text{s.t.}\quad &\mathbf{\Omega}^{(v)}\odot(\mathbf{Z}^{(v)}-\mathbf{X}^{(v)})=\mathbf{0}, \mathbf{W}\mathbf{W}^T=\mathbf{I}. \\ &\mathbf{F}\mathbf{F}^T=\mathbf{I}, \mathbf{S}\mathbf{1}=\mathbf{1}, \mathbf{S}\geq\mathbf{0}.\end{aligned} \tag{11}$$

We update Eq. (11) with the alternating iteration strategy.

Fix $\mathbf{W}$, $\mathbf{U}^{(v)}$, $\mathbf{S}$, $\mathbf{Z}^{(v)}$ and $\mathbf{Y}$, update $\mathbf{F}$, the problem (11) is transformed to

$$\min_{\mathbf{F}}\quad tr(\mathbf{F}\mathbf{L}_\mathbf{S}\mathbf{F}^T)\quad \text{s.t.}\quad \mathbf{F}\mathbf{F}^T=\mathbf{I}. \tag{12}$$

According to [41], the optimal solution to $\mathbf{F}$ is composed by the eigenvectors corresponding to the c smallest eigenvalues of $\mathbf{L}_\mathbf{S}$.

Fix $\mathbf{W}$, $\mathbf{F}$, $\mathbf{S}$, $\mathbf{Z}^{(v)}$ and $\mathbf{Y}$, update $\mathbf{U}^{(v)}$, the problem in (11) is transformed to

$$\min_{\mathbf{U}^{(v)}}\quad \left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2. \tag{13}$$

Take the derivative of Eq. (13) and set the derivative equal to 0, then the optimal solution of $\mathbf{U}^{(v)}$ is

$$\mathbf{U}^{(v)}=\mathbf{Z}^{(v)}\mathbf{S}\mathbf{W}^T(\mathbf{W}\mathbf{W}^T)^{-1}. \tag{14}$$

If $\mathbf{W}\mathbf{W}^T$ is irreversible, a small identity matrix is introduced to make sure $\mathbf{W}\mathbf{W}^T$ is reversible.

Fix $\mathbf{F}$, $\mathbf{U}^{(v)}$, $\mathbf{S}$, $\mathbf{W}$ and $\mathbf{Y}$, update $\mathbf{Z}^{(v)}$, the problem in (11) is transformed to

$$\begin{aligned}\min_{\mathbf{Z}^{(v)}}\quad &\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2 \\ \text{s.t.}\quad &\mathbf{\Omega}^{(v)}\odot(\mathbf{Z}^{(v)}-\mathbf{X}^{(v)})=\mathbf{0}.\end{aligned} \tag{15}$$

Take the derivative of Eq. (15) and set the derivative equal to 0, then the optimal solution of $\mathbf{Z}^{(v)}$ is

$$\mathbf{Z}^{(v)}=\mathbf{U}^{(v)}\mathbf{W}\mathbf{S}(\mathbf{S}^T\mathbf{S})^{-1}+\mathbf{\Omega}^{(v)}\odot(\mathbf{X}^{(v)}-\mathbf{U}^{(v)}\mathbf{W}\mathbf{S}(\mathbf{S}^T\mathbf{S})^{-1}). \tag{16}$$

Fix $\mathbf{F}$, $\mathbf{U}^{(v)}$, $\mathbf{W}$, $\mathbf{Z}^{(v)}$ and $\mathbf{Y}$, update $\mathbf{S}$, the problem in (11) can be written as

$$\begin{aligned}\min_{\mathbf{S}} &\sum_{v=1}^{V}\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2+\alpha tr(\mathbf{W}\mathbf{L}_\mathbf{S}\mathbf{W}^T) \\ &+\beta tr(\mathbf{Y}^T\mathbf{L}_\mathbf{S}\mathbf{Y})+\lambda tr(\mathbf{F}\mathbf{L}_\mathbf{S}\mathbf{F}^T) \\ \text{s.t.}\quad &\mathbf{S}\mathbf{1}=\mathbf{1}, \mathbf{S}\geq\mathbf{0}.\end{aligned} \tag{17}$$

The combination of self-expression technology and matrix factorization makes the solution more complicated. In order

to better solve (17), the pseudo inverse of ${\mathbf{Z}^{(v)}}^T\mathbf{Z}^{(v)}$ is introduced, we get

$$\begin{aligned}\min_{\mathbf{S}\in\Phi}\sum_{v=1}^{V}&\left\|\mathbf{S}-({\mathbf{Z}^{(v)}}^T\mathbf{Z}^{(v)})^{-1}{\mathbf{Z}^{(v)}}^T\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2\\&+\alpha tr(\mathbf{W}\mathbf{L}_{\mathbf{S}}\mathbf{W}^T)+\beta tr(\mathbf{Y}^T\mathbf{L}_{\mathbf{S}}\mathbf{Y})+\lambda tr(\mathbf{F}\mathbf{L}_{\mathbf{S}}\mathbf{F}^T)\\\text{s.t.}\quad&\mathbf{S}\mathbf{1}=\mathbf{1},\mathbf{S}\geq\mathbf{0}.\end{aligned}\tag{18}$$

According to **Theorem 2** proved in next section, solving problem in (17) is equivalent to solving (18).

In spectral analysis, there is a fundamental but important equation

$$\sum_{i,j}^{n}\|\mathbf{f}_i-\mathbf{f}_j\|_2^2 s_{i,j}=2tr(\mathbf{F}\mathbf{L}_{\mathbf{S}}\mathbf{F}^T).\tag{19}$$

Combining Eq. (18) and Eq. (19), we have

$$\begin{aligned}\min_{\mathbf{S}}\sum_{v=1}^{V}&\left\|\mathbf{S}-({\mathbf{Z}^{(v)}}^T\mathbf{Z}^{(v)})^{-1}{\mathbf{Z}^{(v)}}^T\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2\\&+\frac{\alpha}{2}\sum_{i,j}^{n}\|\mathbf{w}_i-\mathbf{w}_j\|_2^2 s_{i,j}+\frac{\beta}{2}\sum_{i,j}^{n}\|\mathbf{y}_i-\mathbf{y}_j\|_2^2 s_{i,j}\\&+\frac{\lambda}{2}\sum_{i,j}^{n}\|\mathbf{f}_i-\mathbf{f}_j\|_2^2 s_{i,j}\\\text{s.t.}\quad&\mathbf{S}\mathbf{1}=\mathbf{1},\mathbf{S}\geq\mathbf{0}.\end{aligned}\tag{20}$$

Define

$$\begin{aligned}h_{i,j}&=\frac{\alpha}{2}\|\mathbf{w}_i-\mathbf{w}_j\|_2^2 s_{i,j}+\frac{\beta}{2}\|\mathbf{y}_i-\mathbf{y}_j\|_2^2 s_{i,j}\\&\quad+\frac{\lambda}{2}\|\mathbf{f}_i-\mathbf{f}_j\|_2^2 s_{i,j}\\\mathbf{P}&=\sum_{v=1}^{V}({\mathbf{Z}^{(v)}}^T\mathbf{Z}^{(v)})^{-1}{\mathbf{Z}^{(v)}}^T\mathbf{U}^{(v)}\mathbf{W},\end{aligned}\tag{21}$$

we can rewrite Eq. (20) as

$$\begin{aligned}&\min_{\mathbf{s}_i}\left\|\mathbf{s}_i+(\frac{1}{2}\mathbf{h}_i-\mathbf{p}_i)\right\|_2^2\\&\text{s.t.}\quad\mathbf{s}_i^T\mathbf{1}=1,\mathbf{s}_i\geq\mathbf{0},\\&\forall j,(\mathbf{x}_i,\mathbf{x}_j)\in\mathcal{M},\mathbf{s}_{i,j}=\mathbf{s}_{j,i}=\iota.\end{aligned}\tag{22}$$

Suppose there are m_i samples must link the i-th sample, Eq. (22) becomes

$$\begin{aligned}&\min_{\mathbf{s}_i}\left\|\mathbf{s}_i+(\frac{1}{2}\mathbf{h}_i-\mathbf{p}_i)\right\|_2^2\\&\text{s.t.}\quad\mathbf{s}_i^T\mathbf{1}=1-m_i\iota,\mathbf{s}_i\geq\mathbf{0}.\end{aligned}\tag{23}$$

Problem (23) has a closed from solution. The Lagrangian function of (23) is

$$\mathcal{L}_1(\mathbf{s}_i,\zeta,\boldsymbol{\delta}_i)=\left\|\mathbf{s}_i+(\frac{1}{2}\mathbf{h}_i-\mathbf{p}_i)\right\|_2^2-\zeta(\mathbf{s}_i\mathbf{1}-1+m_i\iota)-\boldsymbol{\delta}_i^T\mathbf{s}_i,\tag{24}$$

where ζ and $\boldsymbol{\delta}_i$ Lagrange multipliers. According to the KKT conditions, the optimal solution of $\mathbf{s}_i$ is

$$s_{i,j}=(\mathbf{p}_i-\frac{1}{2}\mathbf{h}_i+\zeta)_+,\tag{25}$$

where $x_+=\max(x,0)$. Define $\mathbf{o}_i=\mathbf{p}_i-\frac{1}{2}\mathbf{h}_i$. Suppose $o_{i,1}$,$o_{i,1}$, $o_{i,2},\ldots,o_{i,n}$ are arranged from small to large. A sparse $\mathbf{s}_i$ with c nonzero values is learned. Therefore, we get

$$\begin{cases}\mathbf{o}_{i,c}+\zeta>0,\\\mathbf{o}_{i,c+1}+\zeta\leq 0.\end{cases}\tag{26}$$

Combining Eq. (25) with the constraint $\mathbf{s}_i\mathbf{1}=1-m_i\iota$, we obtain

$$\zeta=\frac{1-m_i}{c}-\sum_{j=1}^{c}o_{i,j}\tag{27}$$

Therefore, the solution of $s_{i,j}$ is

$$s_{i,j}=o_{i,j}+\frac{1-m_i}{c}-\sum_{j=1}^{c}o_{i,j}.\tag{28}$$

Fix F, $\mathbf{U}^{(v)}$, S, $\mathbf{Z}^{(v)}$ and Y, update W, we get

$$\begin{aligned}\min_{\mathbf{W}}\sum_{v=1}^{V}&\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2+\alpha tr(\mathbf{W}\mathbf{L}_{\mathbf{S}}\mathbf{W}^T)\\\text{s.t.}\quad&\mathbf{W}\mathbf{W}^T=\mathbf{I}.\end{aligned}\tag{29}$$

For problem (29), we solve it by the Curvilinear Search Algorithm in [29].

Fix F, $\mathbf{U}^{(v)}$, S, $\mathbf{Z}^{(v)}$ and W, update Y, since every column of $\mathbf{Y}$ is independent, we can compute $\mathbf{Y}$ by solving

$$\begin{aligned}\min_{\mathbf{y}_g}\quad&tr({\mathbf{y}_g}^T\mathbf{L}_{\mathbf{S}}\mathbf{y}_g),\\\text{s.t.}\quad&\forall(\mathbf{x}_{i_g},\mathbf{x}_{j_g})\in\mathcal{C}:\mathbf{y}_g(i_g)=1,\mathbf{y}_g(j_g)=-1.\end{aligned}\tag{30}$$

To solve $\mathbf{y}_g$, we first rearrange all the samples as $\mathcal{X}_g=\{\mathbf{x}_{i_g},\mathbf{x}_{j_g},\mathbf{x}_1,\cdots,\mathbf{x}_{i_g-1},\mathbf{x}_{i_g+1},\cdots,\mathbf{x}_{j_g-1},\mathbf{x}_{j_g+1},\cdots,\mathbf{x}_n\}$. We rewrite $\mathbf{y}_g=[\mathbf{y}_{l_g}^T,\mathbf{y}_{u_g}^T]^T\in\mathbb{R}^n$, where $\mathbf{y}_{l_g}=[1,-1]^T$. Meanwhile, the Laplacian matrix $\mathbf{L}_{\mathbf{S}_g}$ of the similarity graph $\mathbf{S}_g$ is rearranged accordingly and $\mathbf{L}_{\mathbf{S}_g}=\begin{bmatrix}\mathbf{L}_{l_gl_g}&\mathbf{L}_{l_gu_g}\\\mathbf{L}_{u_gl_g}&\mathbf{L}_{u_gu_g}\end{bmatrix}$ [14], [30]. Therefore, the problem (30) can be rewritten as

$$\begin{aligned}\min_{\mathbf{y}_g}\quad&tr(\mathbf{y}_g^T\mathbf{L}_{\mathbf{S}}\mathbf{y}_g)\\\text{s.t.}\quad&\mathbf{y}_{l_g}=[1,-1]^T.\end{aligned}\tag{31}$$

According to the label propagation theory [30], the problem (30) has a close-form solution. That is

$$\mathbf{y}_{u_g}=-{\mathbf{L}_{uu_g}}^{-1}\mathbf{L}_{ul_g}\mathbf{y}_{l_g}.\tag{32}$$

B. Technical Analysis and Time Complexity

First, we present the content and proof of **Theorem 2**.

Theorem 2. *After introducing the pseudo inverse of ${\mathbf{Z}^{(v)}}^T\mathbf{Z}^{(v)}$, problem (17) and (18) has the same solution.*

Proof. Since problem (17) and (18) are different only in the first term. To proof problem (17) and (18) has the same solution, the idea is to prove that their first term have the same solution. Define

$$\mathcal{L}_2(\mathbf{S})=\left\|\mathbf{Z}^{(v)}\mathbf{S}-\mathbf{U}^{(v)}\mathbf{W}\right\|_F^2,\tag{33}$$

Algorithm 1 CWSC

Input:

1. The data matrix $\mathbf{X} = [\mathbf{X}^{(1)}, \cdots, \mathbf{X}^{(V)}]$,
2. The indicator matrix $[\mathbf{\Omega}^1, \cdots, \mathbf{\Omega}^{(V)}]$,
3. Parameters α, β.

Output:

1.The Similar matrix $\mathbf{S}$ with exactly c connected components.

Initialization:

1. Randomly initialize $\mathbf{U}^{(v)}$, $\mathbf{W}$.
2. Initialize $\mathbf{F}$ by the c eigenvectors of $\mathbf{L_S}$ corresponding to the c smallest eigenvalues.
3. $\mathbf{Y} \in \mathbb{R}^{n \times g_n}$, with a random matrix range in (-1,1). Let $y_g(i_g) = 1$, $y_g(j_g) = -1$ for g-th pair of cannot-linked samples, $\forall(\mathbf{x}_{i_g}, \mathbf{x}_{j_g}) \in \mathcal{C}$

Procedure:

Repeat

1. Update $\mathbf{U}^{(v)}$ according to (14).
2. Compute $\mathbf{Z}^{(v)}$ by (16).
3. Calculate the i-row of $\mathbf{S}$ by solving formula (23).
4. Compute $\mathbf{F}$ by the c eigenvectors of $\mathbf{L_S}$ corresponding to the c smallest eigenvalues.
5. Update $\mathbf{W}$ by the Curvilinear Search Algorithm [29].

Until convergence

the derivatives of (33) is

$$\frac{\partial \mathcal{L}_2}{\partial \mathbf{S}} = 2\mathbf{Z}^{(v)T}(\mathbf{Z}^{(v)}\mathbf{S} - \mathbf{U}^{(v)}\mathbf{W}). \tag{34}$$

Let $\frac{\partial \mathcal{L}_2}{\partial \mathbf{S}} = 0$, the solution of $\mathbf{S}$ is

$$\mathbf{S} = \sum_{v=1}^{V} ({\mathbf{Z}^{(v)}}^T \mathbf{Z}^{(v)})^{-1} {\mathbf{Z}^{(v)}}^T \mathbf{U}^{(v)} \mathbf{W}. \tag{35}$$

Similarly, we define

$$\mathcal{L}_3(\mathbf{S}) = \left\| \mathbf{S} - \sum_{v=1}^{V} ({\mathbf{Z}^{(v)}}^T \mathbf{Z}^{(v)})^{-1} {\mathbf{Z}^{(v)}}^T \mathbf{U}^{(v)} \mathbf{W} \right\|_F^2. \tag{36}$$

The derivative of (36) is

$$\frac{\partial \mathcal{L}_3}{\partial \mathbf{S}} = 2(\mathbf{S} - \sum_{v=1}^{V} ({\mathbf{Z}^{(v)}}^T \mathbf{Z}^{(v)})^{-1} {\mathbf{Z}^{(v)}}^T \mathbf{U}^{(v)} \mathbf{W}). \tag{37}$$

Let $\frac{\partial \mathcal{L}_3}{\partial \mathbf{S}} = 0$, we have

$$\mathbf{S} = \sum_{v=1}^{V} ({\mathbf{Z}^{(v)}}^T \mathbf{Z}^{(v)})^{-1} {\mathbf{Z}^{(v)}}^T \mathbf{U}^{(v)} \mathbf{W}. \tag{38}$$

Since (35) equals (38), **Theorem 2** is proved. ∎

Now, we analyze the time complexity of the algorithm. In this paper, there are five groups of variables to be optimized. We analyze the required complexity to update each variable one by one. When update $\mathbf{U}^{(v)}$, from (14) we know that $\mathbf{U}^{(v)} = \mathbf{Z}^{(v)}\mathbf{S}\mathbf{W}^T(\mathbf{W}\mathbf{W}^T)^{-1}$, which costs $\mathcal{O}(rn^2 + r^2n + +d^{(v)}rn + r^3)$. Thus, solving $\{\mathbf{U}^{(v)}\}_{v=1}^V$ consumes $\mathcal{O}(rn^2 + r^2n + +drn + r^3)$. To solve $\mathbf{F}$, we need to calculate the eigenvalues of $\mathbf{L_S}$, the complexity is $\mathcal{O}(n^3)$. According to (16), the time complexity of calculating $\{\mathbf{Z}^{(v)}\}_{v=1}^V$ is $\mathcal{O}(n^3 + drn + dn)$. From [29], we know the complexity of computing $\mathbf{W}$ is $\mathcal{O}(t_0(r^2n + rn))$, where t_0 is iterations times. To optimize $\mathbf{S}$, the main computation comes from the calculation of $\mathbf{P}$, and the time complexity is $\mathcal{O}(n^3 + dn^2 + r^2 + drn)$. According to Eq.(32), computing $\mathbf{Y}$ needs $\mathcal{O}(g_n((n-2)^3 + (n-2)^2))$. Suppose the number of iterations of Algorithm 1 is t, then the total time complexity of CWSC is $\mathcal{O}(t(3n^3 + (d+2r)n^2 + g_n(n-2)^3 + g_n(n-2)^2 + (t_0r^2 + t_0r + 3dr + r^2)n + r^3))$.

V. Experiments

In this section, we first introduce the experimental setting and the datasets. Secondly, we compare the proposed method with related methods to verify the effectiveness of CWSC. Then we evaluate the impact of parameters on our method. The time comparison results are shown in the last part.

To verify the effectiveness of CWSC, we compare it with PVC [15], IMG [17], SRLC [21], MALG [5] and PCOG [14]. More settings are described as follows.

1) Partial multi-View clustering (PVC) is the first work to study IMC problem [15], which uses matrix factorization to learn a common representation to deal with IMC problem. According to the parameter analysis in the original paper, the regularization parameter α of PVC is adjusted in $\{1, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$.
2) Incomplete Multi-modality Grouping (IMG) [17] tends to explore the compact global structure of data when learning the latent subspace. In IMG, considering that there are three parameters in the problem, we take the author's advice, γ and λ are fixed to 100, 0.01, then the parameter β is adjusted in the range of $\{10^{-1}, 10^{-2}, 10^{-3}\}$.
3) Simultaneous Representation Learning and Clustering for incomplete multi-view data (SRLC) [21]. In SRLC, there are two parameters, according to the parameter learning conclusion in the authors' original article, λ is turned in $\{10^1, 10^{1.5}, 10^2, 10^{2.5}, 10^3\}$, γ is turned in $\{1.1, 1.3, 1.5, 1.7, 1.9\}$.
4) Multi-view clustering with Adaptively Learned Graph (MALG) [5]. MALG is originally a strict unsupervised method for complete multi-view data. To make it applicable to the case with both partial instances and pairwise constraints, the missing parts of the dataset are populated with the mean value. Then it is modified by employing the conclusion of Theorem 1, so that it can use pairwise constraints. For concise presentation, we still denote the modified version as MALG. In MALG, the parameter λ is set view-by-view, $\lambda^{(v)} = \Gamma^{(v)} + \log(({\Gamma^{(v)}}^2 + 1)t_a)$, where t_a is the number of iterations, $\Gamma^{(v)}$ is the median of the losses of all similarities on the v-th view.
5) Semi-supervised clustering via Pairwise Constrained Optimal Graph (PCOG) [14]. PCOG is a single-view clustering method for complete data with pairwise constraints. To make it comparable, we first fill the missing instances with the mean value and then concatenate all views into a big feature matrix, on which PCOG is implemented. In PCOG, parameter λ can be adjusted automatically, we turn parameter γ in the range of $\{10^{-2}, 10^{-1}, 1\}$.

It can be seem that PVC, IMG and SRLC are typical weakly structure-supervised methods, where the first two are matrix factorization based and the last one is graph based. MALG and PCOG are modified to be compatible with both kinds of weak supervision information.

Since initialization is very important in graph-based clustering. To get a better initialization, we first pick out complete samples to construct the similarity matrix on each view. Then, for each pair of instances, the average over its non-zeros similarity values is used as the final similarity. In this way, we obtain the initial similarity matrix. The other three graph-based methods use the corresponding initialization methods in their respective articles. In our method, there are two parameters to turn, α is turned within $\{0.1, 1, 5, 10, 15, 20\}$, β is turned in $\{0.001, 0.005, 0.01, 0.05, 0.1\}$.

A. Dataset Description and Experimental Setup

We conduct experiments on seven benchmark datasets (Yale, Kodak, Ixmas, ORL, Prokaryotic, USPS and Forest) to illustrate the superior performance of CSWC over several existing state-of-the-art approaches.

Yale contains 165 images of 15 volunteers and everyone has fifteen images. This dataset is a gray-scale dataset and mainly includes changes in lighting, facial expressions and posture. To analyze data effectively, five visual features are extracted, i.e., SIFT, HOG, LBP, wavelet texture and GIST.

Kodak is a well-known video dataset in the consumer domain, and all samples come from actual users. It includes 2172 samples divided into 21 categories. Edged direction histogram (EDH), Gabor (GBR), and grid color moment are extracted.

Ixmas is a multi-view action dataset that contains recorded video sequences from five views: back, left, front, right and top [31]. It collects data on ten people doing 12 actions. These 12 actions include Check Watch, Cross Arms, Sit Down, Get Up and Turnaround actions. 360 samples are used in our paper, Every action includes 30 instances.

ORL contains 400 face images of 40 different subjects. Each subject has 10 images. The invariant ranklet feature and the granulometric feature are extracted for experiments.

Prokaryotic includes of 551 prokaryotic samples belonging to 4 classes. It contains a textual view and 2 genomic views [32]. The textual descriptions are summarized into a document-term matrix that records the TF-IDF [33] reweighted word frequencies. The genomic views are the proteome composition and the gene repertoire.

USPS contains 9298 instances of ten handwritten digits (0 to 9). The size of each instance is 1616. In our experiments, 256 dimensional LBP and 32 dimensional Gabor Texture features are extracted.

Forest contains 524 samples belonging to four categories. The dataset is committed to using ASTER satellite images to identify different forests. The features are composed of 9 spectral features and 18 geographically weighted similarity variables [34].

The incomplete multi-view data is obtained by deleting part of the complete dataset. Specifically, 10% to 50% (in steps of 10%) of the samples are randomly selected, then we randomly delete a view from the selected p (p is the missing ratio, $10\% \leq p \leq 50\%$) samples.

For large datasets, utilizing all pairwise constraints is inappropriate in our method. Therefore, it is significant to select the key pairwise constraints. In our experiments, we assume the graph similarity $\mathbf{S}$ is k-sparse. That is, each sample is only connected to its k nearest neighbors in the feature space. To obtain key pairwise constraints, we follow the rules in [14]: the must-link constraints are searched from the "non-neighbors" of the sample, and the cannot-link constraints are obtained by querying the "neighbors" of the sample. In this paper, we take 5% of the searched cannot-link pairs. The number of must-link pairs is 3 times that of cannot-link. But for USPS, since this dataset is too large and the variable $\mathbf{Y}$ is updated column by column. We take 1% of the searched cannot-link pairs, the number of must-link pairs is 15 times that of cannot-link. Compared with selecting pairwise constraints randomly, the active strategy stated above is more efficient. To keep comparison fairness, the same pairwise constraints are used for MALG, PCOG and CWSC.

In this paper, the normalized mutual information (NMI) and the clustering accuracy (ACC) are adopted as evaluation indexes. Since both PVC [15] and IMG [17] need to perform extra k-means clustering to obtain the final clustering partition, we repeat k-means 20 times and adopt the mean value as the result for each independent repetition. Finally, to avoid the unfairness brought by randomness, we show the average of 10 independent experiments. But for the large USPS, we display the average of 5 independent experiments.

B. Clustering Results

The experimental results of the above seven datasets are shown in Table II and Table III.

Through the experimental results in the tables, we have the following findings. (1) On most datasets, the results of NMI and ACC decrease when the missing ratio p increases, which is in according with our intuition. (2) For almost all data sets, our method improves the clustering ACC and NMI by more than 5 percentage points in most of the missing proportions, especially on the USPS dataset, the clustering indexes are improved by more than ten percentage points. (3) On the Prokaryotic dataset, the ACC results of almost all methods are not bad. As for NMI, except for CWSC and IMG, other methods perform poorly. It shows that CWSC has a stronger ability to obtain the clustering structure of the data. It can be owing to that CWSC combines the advantages of the graph-based IMC methods and the matrix factorization based IMC methods. (4) In the Ixmas dataset, the performances of almost all methods are mediocre, and with the increase of the missing ratio, the clustering effect does not significantly decrease. This means that the dataset is relatively complex and difficult to clustering. However, the performances of MALG and our method with the pairwise constraints are significantly better, indicating that the pairwise constraints have a significant effect on this type of dataset. (5) In most cases, CWSC, PCOG and MALG perform better than IMG, PVC and SRLC,

TABLE II: the ACC values (mean (std)) of the seven datasets as the p varies, the best results are highlighted in bold.

Datasets	p	PVC	IMG	SRLC	MALG	PCOG	CWSC
ORL	0.1	0.282(0.036)	0.478(0.037)	0.398(0.022)	0.497(0.043)	0.551(0.032)	**0.619(0.024)**
	0.2	0.274(0.045)	0.435(0.011)	0.390(0.014)	0.461(0.037)	0.472(0.020)	**0.568(0.032)**
	0.3	0.269(0.020)	0.396(0.008)	0.391(0.022)	0.460(0.041)	0.398(0.030)	**0.483(0.053)**
	0.4	0.227(0.036)	0.374(0.010)	0.394(0.033)	0.438(0.032)	0.332(0.016)	**0.528(0.059)**
	0.5	0.218(0.022)	0.356(0.011)	0.379(0.022)	0.428(0.033)	0.281(0.016)	**0.497(0.044)**
Prokaryotic	0.1	0.494(0.058)	0.612(0.010)	0.614(0.065)	0.588(0.034)	0.636(0.051)	**0.725(0.033)**
	0.2	0.483(0.050)	0.612(0.010)	0.614(0.065)	0.607(0.033)	0.593(0.042)	**0.720(0.060)**
	0.3	0.433(0.032)	0.605(0.023)	0.501(0.039)	0.613(0.069)	0.601(0.066)	**0.738(0.069)**
	0.4	0.417(0.039)	0.602(0.019)	0.509(0.073)	0.596(0.038)	0.577(0.041)	**0.723(0.074)**
	0.5	0.392(0.049)	0.582(0.026)	0.469(0.067)	0.613(0.038)	0.575(0.015)	**0.699(0.040)**
Yale	0.1	0.591(0.013)	0.582(0.012)	0.562(0.031)	0.772(0.075)	0.768(0.042)	**0.838(0.055)**
	0.2	0.583(0.014)	0.578(0.013)	0.572(0.029)	0.761(0.049)	0.750(0.025)	**0.820(0.041)**
	0.3	0.550(0.026)	0.567(0.017)	0.574(0.025)	0.712(0.034)	0.764(0.031)	**0.843(0.036)**
	0.4	0.525(0.022)	0.574(0.015)	0.568(0.044)	0.711(0.013)	0.714(0.035)	**0.818(0.031)**
	0.5	0.508(0.068)	0.568(0.012)	0.575(0.030)	0.711(0.098)	0.707(0.048)	**0.764(0.027)**
Ixmas	0.1	0.258(0.011)	0.279(0.005)	0.257(0.012)	0.392(0.069)	0.219(0.011)	**0.465(0.037)**
	0.2	0.260(0.007)	0.281(0.007)	0.253(0.013)	0.380(0.020)	0.189(0.028)	**0.459(0.029)**
	0.3	0.254(0.010)	0.278(0.005)	0.256(0.012)	0.369(0.021)	0.219(0.014)	**0.439(0.041)**
	0.4	0.260(0.015)	0.278(0.007)	0.257(0.013)	0.352(0.026)	0.227(0.010)	**0.438(0.024)**
	0.5	0.254(0.011)	0.275(0.003)	0.261(0.019)	0.338(0.026)	0.214(0.024)	**0.439(0.029)**
Kodak	0.1	0.204(0.007)	0.214(0.003)	0.184(0.007)	0.353(0.014)	0.305(0.013)	**0.376(0.015)**
	0.2	0.199(0.010)	0.208(0.004)	0.188(0.012)	0.346(0.012)	0.288(0.016)	**0.375(0.009)**
	0.3	0.190(0.006)	0.204(0.004)	0.184(0.005)	0.337(0.011)	0.301(0.013)	**0.376(0.013)**
	0.4	0.185(0.006)	0.200(0.004)	0.185(0.011)	0.306(0.048)	0.293(0.017)	**0.376(0.013)**
	0.5	0.182(0.007)	0.201(0.006)	0.185(0.010)	0.305(0.006)	0.260(0.034)	**0.386(0.008)**
USPS	0.1	0.404(0.010)	0.426(0.013)	0.424(0.017)	0.501(0.015)	0.492(0.026)	**0.714(0.026)**
	0.2	0.389(0.011)	0.405(0.014)	0.429(0.019)	0.486(0.025)	0.461(0.016)	**0.638(0.051)**
	0.3	0.356(0.016)	0.374(0.009)	0.396(0.025)	0.434(0.022)	0.401(0.012)	**0.626(0.033)**
	0.4	0.315(0.016)	0.350(0.014)	0.405(0.031)	0.431(0.035)	0.413(0.027)	**0.655(0.045)**
	0.5	0.282(0.037)	0.301(0.012)	0.374(0.027)	0.365(0.046)	0.350(0.054)	**0.573(0.048)**
Forest	0.1	0.620(0.022)	0.766(0.004)	0.712(0.052)	0.820(0.055)	0.836(0.046)	**0.875(0.036)**
	0.2	0.615(0.036)	0.763(0.013)	0.706(0.078)	0.805(0.064)	0.814(0.063)	**0.855(0.054)**
	0.3	0.587(0.065)	0.760(0.010)	0.708(0.033)	0.787(0.078)	0.795(0.086)	**0.832(0.075)**
	0.4	0.525(0.059)	0.750(0.011)	0.701(0.058)	0.769(0.083)	0.774(0.079)	**0.814(0.081)**
	0.5	0.496(0.037)	0.744(0.010)	0.700(0.049)	0.764(0.091)	0.768(0.090)	**0.812(0.050)**

which shows that the pairwise constraints do help improve the clustering performance. In the above three methods that utilize the weak label supervision information, CWSC performs better than MALG and PCOG, which further shows that CWSC, which combines matrix factorization technology and graph technology, is better than PCOG and MALG that only use graph technology. Above all, it is not difficult to see that combining the advantages of the matrix decomposition based IMC method and the graph-based IMC method, while using the prior information of the pairwise constraints, can help us better deal with different types of datasets.

C. Ablation Experiment

To further illustrate that adding pairwise constraints is meaningful, we compare with the variant of CWSC which does not use pairwise constraints. Since this variant jointly performs Subspace Clustering and Subspace Learning, we name it as SCSL for short. At the same time, we compare SCSL with all the IMC methods without pairwise constraints to show that our proposed CWSC is effective in dealing with weak structure supervision information without any weak label supervision information. The experimental results are shown in Fig. 2 and Fig. 3.

From Fig 2 and Fig 3, we find that even without using pairwise constraints, SCSL is still better than traditional graph-based and matrix factorization based IMC methods on the three datasets Yale, Forest and Kodak. These results directly show that it is meaningful to integrate the two mainstream methods through the self-expression technique. In addition, through the comparison between SCSL and CWSC, we learn that the rational use of weak label supervision information does improve clustering performance.

D. Discussion on the Number of Pairwise Constraints

In our method, pairwise constraints play a significant role in clustering. To further exploit the relationship between the number of pairwise constraints and the clustering performance, we make experiments on Yale, ORL and Ixmas datasets when the missing ratio is 0.2. Specifically, we set the number of cannot-link pairs from 50 to 300 (the step is 50), and the number of the must-link constraints is three times that of the

TABLE III: the NMI values (mean (std)) of the seven datasets as the p varies, the best results are highlighted in bold.

Datasets	p	PVC	IMG	SRLC	MALG	PCOG	CWSC
ORL	0.1	0.533(0.042)	0.712(0.018)	0.643(0.011)	0.705(0.041)	0.733(0.012)	**0.785(0.015)**
	0.2	0.528(0.055)	0.672(0.006)	0.638(0.013)	0.671(0.038)	0.654(0.014)	**0.736(0.024)**
	0.3	0.522(0.019)	0.632(0.008)	0.637(0.019)	0.660(0.042)	0.577(0.018)	**0.665(0.034)**
	0.4	0.467(0.048)	0.612(0.011)	0.637(0.029)	0.629(0.033)	0.525(0.009)	**0.684(0.042)**
	0.5	0.454(0.030)	0.591(0.007)	0.632(0.022)	0.616(0.033)	0.472(0.015)	**0.657(0.035)**
Prokaryotic	0.1	0.213(0.079)	0.380(0.012)	0.356(0.078)	0.154(0.089)	0.292(0.075)	**0.409(0.063)**
	0.2	0.186(0.053)	0.380(0.012)	0.356(0.078)	0.196(0.088)	0.130(0.118)	**0.397(0.076)**
	0.3	0.139(0.030)	0.375(0.026)	0.237(0.038)	0.202(0.128)	0.157(0.167)	**0.411(0.075)**
	0.4	0.108(0.033)	0.365(0.019)	0.245(0.071)	0.160(0.087)	0.078(0.102)	**0.383(0.090)**
	0.5	0.088(0.040)	**0.344(0.026)**	0.203(0.053)	0.188(0.085)	0.089(0.058)	0.336(0.056)
Yale	0.1	0.647(0.012)	0.652(0.008)	0.585(0.021)	0.771(0.047)	0.669(0.027)	**0.882(0.039)**
	0.2	0.648(0.018)	0.647(0.010)	0.597(0.026)	0.776(0.068)	0.618(0.020)	**0.865(0.027)**
	0.3	0.616(0.025)	0.640(0.012)	0.595(0.022)	0.731(0.012)	0.529(0.034)	**0.842(0.023)**
	0.4	0.600(0.020)	0.646(0.014)	0.588(0.040)	0.730(0.028)	0.481(0.032)	**0.819(0.023)**
	0.5	0.569(0.065)	0.637(0.012)	0.595(0.029)	0.724(0.085)	0.399(0.041)	**0.788(0.023)**
Ixmas	0.1	0.261(0.011)	0.324(0.004)	0.291(0.011)	0.500(0.086)	0.276(0.047)	**0.548(0.027)**
	0.2	0.268(0.017)	0.325(0.007)	0.295(0.022)	0.493(0.042)	0.254(0.088)	**0.541(0.023)**
	0.3	0.252(0.016)	0.326(0.005)	0.291(0.012)	0.468(0.070)	0.307(0.012)	**0.526(0.027)**
	0.4	0.257(0.016)	0.323(0.005)	0.290(0.011)	0.446(0.072)	0.307(0.025)	**0.523(0.023)**
	0.5	0.289(0.032)	0.320(0.004)	0.292(0.014)	0.412(0.072)	0.273(0.051)	**0.525(0.030)**
Kodak	0.1	0.210(0.009)	0.225(0.002)	0.180(0.007)	0.199(0.019)	0.103(0.020)	**0.235(0.027)**
	0.2	0.205(0.007)	0.216(0.003)	0.178(0.009)	0.182(0.021)	0.081(0.012)	**0.238(0.011)**
	0.3	0.197(0.008)	0.208(0.004)	0.178(0.009)	0.162(0.019)	0.096(0.020)	**0.237(0.022)**
	0.4	0.184(0.007)	0.200(0.004)	0.174(0.012)	0.143(0.024)	0.077(0.018)	**0.237(0.022)**
	0.5	0.179(0.007)	0.195(0.005)	0.179(0.011)	0.141(0.011)	0.070(0.016)	**0.230(0.013)**
USPS	0.1	0.487(0.009)	0.467(0.011)	0.459(0.020)	0.456(0.025)	0.443(0.027)	**0.632(0.026)**
	0.2	0.468(0.023)	0.452(0.013)	0.461(0.024)	0.439(0.017)	0.428(0.041)	**0.554(0.047)**
	0.3	0.447(0.008)	0.436(0.010)	0.448(0.019)	0.425(0.029)	0.425(0.034)	**0.546(0.054)**
	0.4	0.426(0.020)	0.405(0.014)	0.432(0.026)	0.413(0.044)	0.377(0.038)	**0.522(0.058)**
	0.5	0.357(0.018)	0.350(0.025)	0.416(0.030)	0.342(0.031)	0.314(0.046)	**0.430(0.043)**
Forest	0.1	0.452(0.023)	0.512(0.006)	0.485(0.011)	0.626(0.036)	0.632(0.077)	**0.687(0.022)**
	0.2	0.443(0.054)	0.513(0.012)	0.502(0.045)	0.599(0.063)	0.600(0.087)	**0.661(0.039)**
	0.3	0.396(0.082)	0.503(0.014)	0.492(0.031)	0.581(0.092)	0.586(0.084)	**0.645(0.049)**
	0.4	0.371(0.047)	0.489(0.017)	0.489(0.031)	0.550(0.069)	0.556(0.080)	**0.627(0.064)**
	0.5	0.334(0.051)	0.478(0.012)	0.487(0.027)	0.545(0.086)	0.549(0.067)	**0.615(0.062)**

TABLE IV: The time consumption of different methods, the best results are highlighted in bold

Data	Methods					
	PVC	IMG	SRLC	MALG	PCOG	WCSC
Yale	**0.1113**	2.1137	0.3039	0.0233	0.4106	1.5362
Prokaryotic	**0.0905**	19.404	0.3816	0.4885	3.6724	3.1314
USPS	**3.6594**	7736.5	90.629	463.86	549.55	559.52

cannot-link, and the experiments are carried out separately. The experiment results are shown in Fig. 4.

It can be seen from Fig. 4 that with the increase of the pairwise constraints, the effects of the clustering indicators ACC and NMI on the three datasets gradually increase. Intuitively, the more pairwise constraints we use, the better clustering results we get. However, in Fig. 4(b), we can see that when the pairwise constraints rise from 200 to 300, ACC basically does not rise, and NMI first rises a bit and then falls, which means blindly pursuing an increase in the number of pairwise constraints is not a good choice for some datasets.

E. Parameter Study

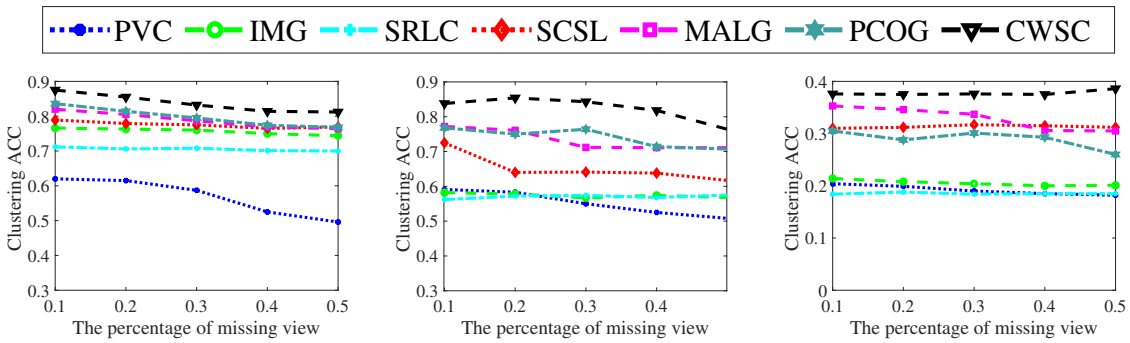
In CWSC, there are two parameters α and β to turn. When the missing ratio p is 0.3, we turn the parameters in Yale, Ixmas and ORL. α is turned within $\{0.1, 1, 5, 10, 15, 20\}$, β is turned in $\{0.001, 0.005, 0.01, 0.05, 0.1\}$. The experimental results are shown in Fig. 5.

As shown in the Fig.5, adjusting α and β has a significant impact on the experimental results. It can be seen from the Fig. 5, α should not be too small, β should not be too large. When α is $\{10,15,20\}$ and β is $\{0.001,0.005\}$, the algorithm effect is relatively stable.

F. Time Consumption

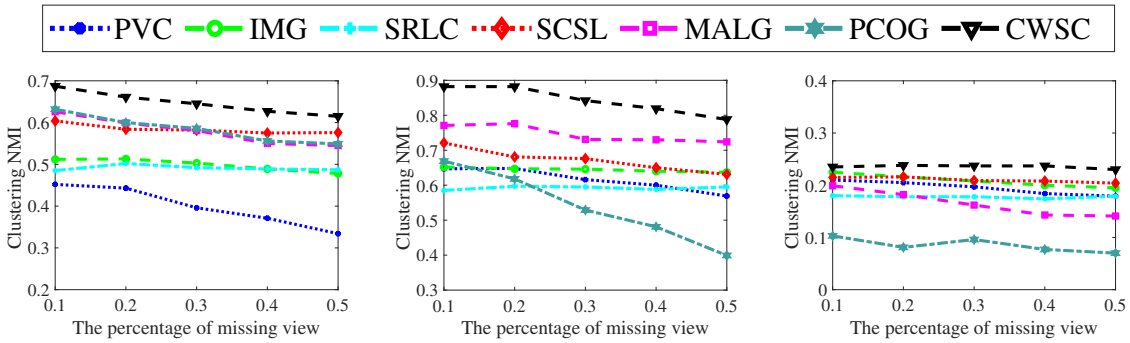
In section IV-B, we theoretically analyze the time complexity of CWSC. In this section, we compare the length of time spent in the actual experiment. When the missing ratio is 0.5, we make experiments on Yale, Prokaryotic, and USPS. The training time is shown in Table IV.

From Table IV, it shows that our method requires a moderate amount of running time on all datasets, and PVC costs



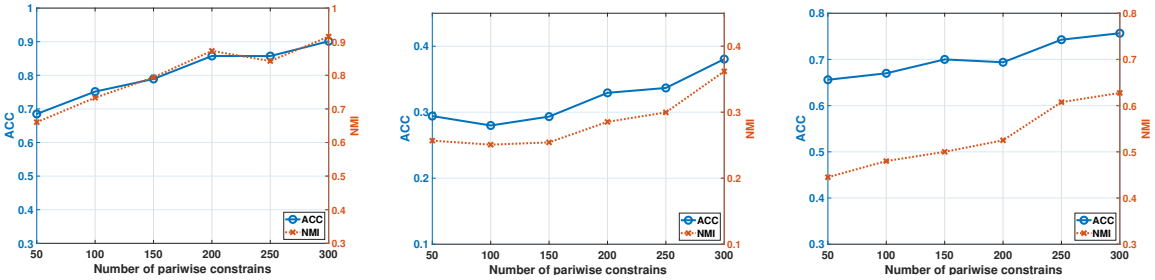
(a) Forest (b) Yale (c) Kodak

Fig. 2: The ACC values (mean (std)) of the three datasets as the incomplete ratio p varies.



(a) Forest (b) Yale (c) Kodak

Fig. 3: The NMI values (mean (std)) of the three datasets as the incomplete ratio p varies.



(a) Yale (b) Ixmas (c) ORL

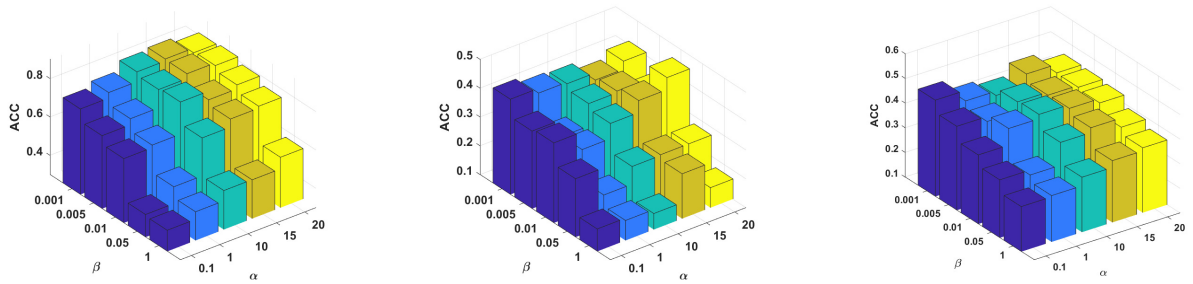
Fig. 4: As the number of pairwise constraints changes, the results of ACC and NMI.

less time than all other methods, while IMG costs much more time especially on big datasets.

VI. Application in Face Clustering

In real life, studying how to cluster the faces that appear in videos can be applied in many fields, including content-based video retrieval, automatic collection of large-scale facial data, etc. In the actual video, the shape of the face that is affected by the lighting conditions and the facial expressions changes significantly. Therefore, it is not a simple case to solve the face clustering problem.

Notting-Hill is a video face dataset [35] derived from the movie Notting-Hill. The faces of five main casts are collected, including 4660 faces in 76 tracks. We separately cluster the faces of three actors, four actors, and five actors, corresponding to data sets Notting-Hill3, Notting-Hill4 and Notting-Hill5. The 110 images are randomly selected of each cast. In Notting-Hill dataset, the pairwise constraints information is naturally available. As the example shown in Fig. 6, two faces of the same track are must-link instances, and the two faces that appear in the same frame cannot be linked. Consistent with the previous experimental settings, we applied CWSC and several other comparison methods to Notting-Hill3, Notting-Hill4 and Notting-Hill5. The experimental results are shown in Table V and Table VI. It can be seen that the overall performance of CWSC is significantly better than other methods.



(a) Yale (b) Ixmas (c) ORL

Fig. 5: When the missing ratio is 0.3, the parameter study of CWSC.

TABLE V: the ACC values (mean (std)) of 3 Notting-Hill datasets as the p varies, the best results are highlighted in bold.

Datasets	p	PVC	IMG	SRLC	MALG	PCOG	CWSC
Notting-Hill3	0.1	0.721(0.062)	0.815(0.013)	0.824(0.053)	0.861(0.087)	0.832(0.097)	**0.931(0.010)**
	0.2	0.697(0.085)	0.802(0.016)	0.831(0.081)	0.843(0.076)	0.818(0.081)	**0.928(0.013)**
	0.3	0.662(0.065)	0.785(0.027)	0.810(0.076)	0.831(0.065)	0.781(0.062)	**0.924(0.009)**
	0.4	0.673(0.062)	0.776(0.018)	0.778(0.074)	0.814(0.054)	0.796(0.072)	**0.930(0.018)**
	0.5	0.614(0.057)	0.754(0.024)	0.769(0.080)	0.809(0.052)	0.774(0.049)	**0.921(0.014)**
Notting-Hill4	0.1	0.692(0.057)	0.743(0.020)	0.742(0.071)	0.832(0.069)	0.761(0.078)	**0.872(0.013)**
	0.2	0.679(0.063)	0.729(0.012)	0.763(0.064)	0.794(0.071)	0.747(0.048)	**0.881(0.017)**
	0.3	0.683(0.049)	0.738(0.027)	0.761(0.065)	0.806(0.061)	0.736(0.068)	**0.876(0.015)**
	0.4	0.653(0.038)	0.727(0.011)	0.724(0.081)	0.783(0.048)	0.721(0.072)	**0.869(0.011)**
	0.5	0.584(0.051)	0.731(0.018)	0.730(0.069)	0.762(0.039)	0.730(0.061)	**0.865(0.018)**
Notting-Hill5	0.1	0.656(0.084)	0.753(0.014)	0.717(0.064)	0.855(0.127)	0.792(0.114)	**0.912(0.015)**
	0.2	0.643(0.098)	0.760(0.014)	0.715(0.093)	0.810(0.085)	0.808(0.060)	**0.927(0.014)**
	0.3	0.623(0.047)	0.759(0.025)	0.765(0.096)	0.829(0.052)	0.781(0.057)	**0.916(0.014)**
	0.4	0.614(0.027)	0.762(0.021)	0.736(0.088)	0.803(0.044)	0.776(0.074)	**0.915(0.016)**
	0.5	0.538(0.073)	0.760(0.022)	0.750(0.066)	0.813(0.047)	0.753(0.051)	**0.921(0.012)**

TABLE VI: the NMI values (mean (std)) of 3 Notting-Hill datasets as the p varies, the best results are highlighted in bold.

Datasets	p	PVC	IMG	SRLC	MALG	PCOG	CWSC
Notting-Hill3	0.1	0.613(0.085)	0.803(0.011)	0.753(0.067)	0.797(0.088)	0.783(0.076)	**0.965(0.012)**
	0.2	0.585(0.081)	0.796(0.019)	0.742(0.094)	0.772(0.076)	0.763(0.061)	**0.959(0.010)**
	0.3	0.553(0.049)	0.785(0.023)	0.731(0.075)	0.773(0.061)	0.758(0.072)	**0.963(0.011)**
	0.4	0.521(0.061)	0.783(0.028)	0.729(0.068)	0.767(0.075)	0.731(0.049)	**0.958(0.017)**
	0.5	0.452(0.060)	0.774(0.034)	0.734(0.053)	0.759(0.059)	0.708(0.052)	**0.960(0.018)**
Notting-Hill4	0.1	0.582(0.084)	0.717(0.017)	0.637(0.084)	0.745(0.075)	0.718(0.074)	**0.915(0.015)**
	0.2	0.551(0.075)	0.725(0.016)	0.643(0.072)	0.738(0.068)	0.706(0.062)	**0.927(0.016)**
	0.3	0.527(0.083)	0.727(0.024)	0.674(0.079)	0.727(0.057)	0.713(0.070)	**0.918(0.012)**
	0.4	0.492(0.068)	0.721(0.021)	0.647(0.068)	0.735(0.064)	0.696(0.058)	**0.920(0.013)**
	0.5	0.446(0.052)	0.706(0.027)	0.653(0.101)	0.724(0.063)	0.702(0.060)	**0.919(0.014)**
Notting-Hill5	0.1	0.531(0.094)	0.729(0.009)	0.637(0.094)	0.760(0.119)	0.739(0.083)	**0.944(0.016)**
	0.2	0.491(0.096)	0.736(0.018)	0.643(0.100)	0.747(0.056)	0.726(0.054)	**0.937(0.014)**
	0.3	0.474(0.063)	0.732(0.025)	0.674(0.095)	0.761(0.064)	0.736(0.065)	**0.947(0.010)**
	0.4	0.464(0.052)	0.730(0.023)	0.647(0.081)	0.752(0.082)	0.696(0.057)	**0.953(0.015)**
	0.5	0.344(0.096)	0.718(0.033)	0.653(0.086)	0.741(0.074)	0.676(0.049)	**0.933(0.015)**

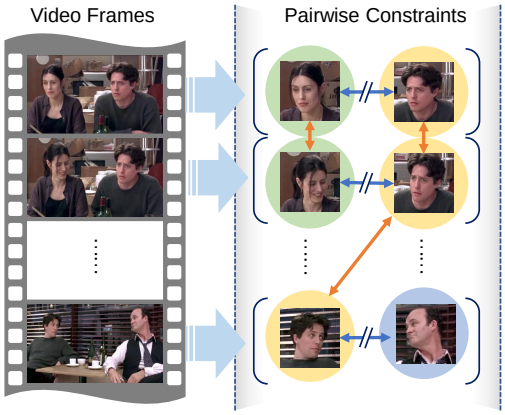


Fig. 6: On the left side of the figure, there are three frames of the movie Notting-Hill, and the pairwise constraint information of the samples is shown on the right side. The must-link samples are linked with an orange arrow, and the cannot-link samples are linked with a cut-off blue arrow.

VII. Conclusion

In this paper, we focus on the problem of improving clustering performance by using weak supervision information. To address this problem, CWSC which effectively utilizes weak supervision information at the data level and feature level is proposed. Specifically, we first convert the pairwise constraints into a graph regular term to better combine with the weak structure supervised model. Then, instead of simply combining the graph-based IMC method with the pairwise constraints, the self-expression technique is introduced to naturally combine the two types of IMC methods. The resultant model can not only effectively integrate the weak label supervision information and weak structure supervision information into a unified model, but also overcome the disadvantages of two types of IMC methods. Finally, by adding a rank constraint, we can obtain the clustering results by directly cutting the learned graph. The effectiveness of CWSC is validated with experimental results of various datasets and the face clustering application.

In further work, we plan to work on reducing the computation cost, so that the clustering efficiency can be improved.

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